

Day 4 – Python Roadmap Notes (Loops)

Today's goal is to master loops, because they're one of the core building blocks of programming and appear constantly in coding interviews, automation scripts, AI preprocessing, and real projects.

What are Loops?

Loops allow you to execute a block of code multiple times without rewriting it.

Instead of writing:

```
print("AI") print("AI") print("AI")  
print("AI") print("AI")
```



Use a loop:



```
for i in range(5): print("AI")
```

Output:



```
AI AI AI AI AI
```

Types of Loops in Python

1. for Loop

Used when you know how many times to repeat.

Syntax



```
for variable in iterable: # code
```

Example



```
for i in range(5): print(i)
```

Output



```
0 1 2 3 4
```

range()

Creates a sequence of numbers.



```
range(5)
```

Produces



```
0 1 2 3 4
```

Examples



```
range(2,8)
```

Output



```
2 3 4 5 6 7
```

Step value



```
range(0,10,2)
```

Output



```
0 2 4 6 8
```

Reverse



```
range(10,0,-1)
```

Output



```
10 9 8 7 6 5 4 3 2 1
```

Iterating through a List



```
languages = ["Python","Java","C++"]  
for lang in languages: print(lang)
```

Output



```
Python Java C++
```

Iterating through a String



```
for letter in "Python": print(letter)
```

Output



```
P y t h o n
```

while Loop

Used when the number of repetitions is unknown.

Syntax



```
while condition: # code
```

Example



```
count = 1 while count <= 5:  
print(count) count += 1
```

Output



```
1 2 3 4 5
```

Infinite Loop



```
while True: print("Hello")
```


Stops only when manually interrupted.

break

Stops the loop immediately.



```
for i in range(10): if i == 5: break print(i)
```

Output



```
0 1 2 3 4
```

continue

Skips the current iteration.



```
for i in range(6):  
    if i == 3: continue  
    print(i)
```

Output



```
0 1 2 4 5
```

pass

Acts as a placeholder when you haven't written

the code yet.



```
for i in range(5): pass
```

Nested Loops

A loop inside another loop.



```
for i in range(3): for j in range(3):  
    print(i, j)
```

Output



```
0 0 0 1 0 2 1 0 1 1 1 2 2 0 2 1 2 2
```

Mini Practice

Print numbers from 1 to 10.



```
for i in range(1,11): print(i)
```

Print even numbers.



```
for i in range(2,21,2): print(i)
```

Print multiplication table of 7.



```
for i in range(1,11): print(f"7 x {i} =  
{7*i}")
```

Find the sum of numbers.



```
total = 0 for i in range(1,6): total += i  
print(total)
```

Output



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Guessing game with while



```
secret = 8 guess = 0 while guess !=  
secret: guess = int(input("Enter  
number: ")) print("Correct!")
```

Interview Questions

Q1. Difference between for and while?

for → Use when the number of iterations is known.

while → Use when iterations depend on a

condition.

Q2. What does break do?

It exits the loop immediately.

Q3. What does continue do?

It skips the current iteration and moves to the next one.

Q4. What is a nested loop?

A loop inside another loop.

Mini Project – Number Guessing Game



```
import random
secret = random.randint(1,20)
while True:
    guess = int(input("Guess: "))
    if guess == secret:
        print("You Win!")
        break
    elif guess < secret:
        print("Too Low")
    else:
        print("Too High")
```

Day 4 Assignment

Learn for loops.

Learn while loops.

Practice range().

Use break, continue, and pass.

Solve 10 loop-based problems on platforms like LeetCode or HackerRank.

Build the Number Guessing Game.

Push today's code to your GitHub repository.

Tomorrow (Day 5): Functions in Python – learn how to write reusable code with parameters, return values, scope, and lambda functions.